

APPROACH TO EXCESSIVE CRYING IN AN INFANT

1. “NORMAL” CRYING PATTERNS

- 0 – 6 weeks old: 3H – 4H per day
- 7 – 9 weeks old: 2H – 3H per day
- 10 – 12 weeks old: 1H – 2H per day

2. CAUSES OF EXCESSIVE CRYING

2.1 COMMON CAUSES

- Ambient temperature (too warm → discomfort from perspiration; Too cold)
- Hungry / overfeeding
- Nasal congestion
- Soiled diapers
- Parental anxiety

2.2 PATHOLOGICAL CAUSES

SYSTEM	CONDITION	CLINICAL FEATURES
CNS	Meningitis	Bulging fontanelle, AMS, fever, vomiting
	Shaken baby syndrome	Retinal hemorrhages, unexplained bruises
CVS	SVT	Pallor, irritability, cool peripheries, poor feeding
	Cardiac failure	Poor perfusion, tachypnea/distress during feeds, poor growth
GI	Inguinal hernia	Bulge in perineal area
	Intussusception	Bilious vomiting, abdominal mass, abdominal distension, AMS
	Anal fissures	Apparent on physical examination
	GERD	Vomiting, irritability when placed supine
GU	Torsion of ovary / testis	Testicular tenderness
	UTI	Vomiting, fever
Skin / MSK	Hair tourniquet	Apparent on physical examination
	Diaper rash	Apparent on physical examination
	Fractures	Asymmetric Moro's, reduced movement of limb
	Osteomyelitis / septic arthritis	Pain on movement of joint, reduced movement of limb, unequal fat folds (reduced folds 2' swelling)
Eye	Corneal abrasion / FB	Conjunctivitis, discharge
General	Drug intoxication	Toxidromes, jittery
	Neonatal abstinence syndrome	Maternal prenatal substance abuse, positive urine screen

3. EVALUATION

- Goal of clinical evaluation is to identify pathological causes of excessive crying, secondary to:
 - Infection
 - Pain (mechanical vs. inflammatory)
 - Noxious stimuli (e.g., palpitations, nausea, reflux, drug withdrawal)
- Red flag = poor weight gain
- A top to toe examination must be carried out, looking for pathological causes of pain (table above)
- Investigations should be based on clinical suspicion & differential diagnosis – routine investigations for screening are not recommended.

4. INFANTILE COLIC

- Benign, self-limiting condition that resolves with time

4.1 DEFINITION

- Crying for no apparent reason, for ≥ 3 H per day
- Occurs ≥ 3 days per week
- Previously healthy infants < 3 months of age

4.2 CLINICAL FEATURES OF INFANTILE COLIC

- Paroxysmal patterns – Clear beginning & end, no relation to what infant was doing prior
Early peak & evening clustering
- Qualitative differences – Louder, higher, & more variable in pitch
More turbulent & dysphonic
Sounds like infant is in pain / or screaming
- Hypertonia – Facial flushing, circum-oral pallor
Tense or distended abdomen, drawing up of legs
Clenching fingers, stiffening & tightening of arms, arching of back
- Difficulty consoling – Difficult to console no matter what parents do

4.3 PROPOSED ETIOLOGIES – CAUSE IS NOT KNOWN

- Gastrointestinal: Aerophagia, altered fetal microflora
- Biologic: Tobacco smoke & nicotine exposure during pregnancy / postpartum period, early form of migraine
- Psychosocial:
 - Parental variables: Family stress, maternal anxiety
Transfer of maternal tension to infant
 - Overstimulation: Discharge after a long day of exposure to environmental stimuli

4.4 MANAGEMENT

4.4.1 PARENTAL SUPPORT IS THE MAINSTAY OF MANAGEMENT

- Resolves spontaneously by 3 – 4 months
- Reassurance – infant is not sick or in pain
- This is not infant's way of rejecting parents, or something they are/are not doing
- Teach first-line interventions
- Encourage breaks from infant
- Acknowledging feelings

4.4.2 FIRST-LINE INTERVENTIONS

- Feeding techniques: Reduce swallowed air by holding bottle at a steeper incline to reduce air-fluid level @ the teat, frequent burping to remove swallowed air.
- Soothing techniques: Pacifier, car-ride, walk on stroller, change scenery (minimize visual stimulation), warm bath, swaddling, "white noise"

4.4.3 ADVICE FOR PARENTS

- Establish a routine
- Alternative care for parental respite

4.4.4 UNPROVEN INTERVENTIONS

- Change in diet:
 - Extensive hydrolysate formulas
 - Soy protein formula
- Probiotics
 - *Lactobacillus reuteri* has inconsistent benefit in reducing crying time
 - Counsel parents regarding:
 - Cost of *L. reuteri*
 - Limited benefit with inconsistent results
 - Natural history of infantile colic
 - No adverse effects or harm reported so far
 - Other probiotics are not recommended
- Simethicone
 - Cause gas bubbles to coalesce, facilitating expulsion
 - Limited benefit with conflicting results
 - Considered safe (may interact with levothyroxine and result in under treatment)

5. INDICATIONS FOR ADMISSION

- Assessment suggests organic illness requiring inpatient care
- Remains inconsolable after a period of observation in CE
- Severe parental anxiety & inability to cope
- Maternal post-natal depression
- Reattendance